GUJARAT TECHNOLOGICAL UNIVERSITY
Chandkheda, Ahmedabad
Affiliated

Government Engineering College,Gandhinagar
A Report On

EMOTION BASED MUSIC PLAYER

Under subject of DESIGN ENGINEERING II-A
B. E. Semester: V
(IT Engineering)

Submitted by:
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GOVERNMENT ENGINEERING COLLEGE, GANDHINAGAR
IT DEPARTMENT

CANDIDATE'S DECLARATION

We hereby declare that the work presented in this project entitled **" Emotion based music player "** submitted towards completion of project in fourth Semester of B.E. (IT) is an authentic record of our original work carried out under the guidance of **"Prof. Anamika mittal "**. We have not submitted the matter embodied in this project for the award of any other degree.

Semester: 5th

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CERTIFICATE

Date:

This is to certify that the **"Emotion based music player"** has been carried out by PATEL BHUMI, GRACY RATHOD, AYUSHI PRAJAPATI, RINKI DEVI under my guidance in fulfillment of the subject Design Engineering in Information Technology Engineering (5 " Semester) of Gujarat Technological University, Ahmadabad during the academic year 2023-24.

Prof. Anamika Mittal — Prof. Jitendrakumar S. Dhobi

(Internal Guide) — **(Head of the Department)**

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We even thank and appreciate to our colleague in developing the project and people who have willingly helped us out with their abilities.

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ABSTRACT

Human expression plays a vital role in determining the current state and mood of an individual, it helps in extracting and understanding the emotion that an individual has based on various features of the face such as eyes, cheeks, forehead or even through the curve of the smile.

Music is basically an art form that soothes and calms human brain and body. Taking these two aspects and blending them together our project deals with detecting emotion of an individual through facial expression and playing music according to the mood detected that will alleviate the mood or simply calm the individual and can also get quicker song according to the mood, saving time from looking up different songs and parallel developing a software that can be used anywhere with the help of providing the functionality of playing music according to the emotion detected.

By developing a recommendation system, it could assist a user to make a decision regarding which music one should listen to helping the user to reduce his/her stress levels. The user would not have to waste any time in searching or to look up for songs and the best track matching the user's mood is detected, and songs would be shown to the user according to his/her mood. The image of the user is captured with the help of a webcam. The user's picture is taken and then as per the mood/emotion of the user an appropriate song from the playlist of the user is shown matching the user's requirement.

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CHEPTER 1

INTRODUTION TO DOMAIN

Music plays a very primary role in elevating an individual ‘s life as it is an important medium of entertainment for music lovers and listeners. In today ‘s world, with the increasing advancements in the field of multimedia and technology, various music players have been developed with features like fast forward, reverse, variable playback speed, genre classification, streaming playback with multicast streams and including volume modulation, etc.

These features might satisfy the user ‘s basic requirements, but the user has got to face the task of manually browsing the playlist of songs and choose songs supported their current mood and behavior. Emotion based music player is a novel approach that helps the user to automatically play songs according to the emotions of the user. It recognizes the facial emotions of the user and plays the songs according to their emotion. The emotions are recognized using a machine learning method EMO algorithm. The human face is an important organ of an individual ‘s body and it especially plays an important role in extraction of an individual ‘s behaviors and emotional state. The webcam captures the image of the user. It then extracts the facial features of the user from the captured image. Facial expression categorized into 2, smiling and not smiling.

The foremost concept of this project is to automatically play songs based on the emotions of the user. It aims to provide user-preferred music with respect to the emotions detected. In existing system user has to manually select the songs, randomly played songs may not match to the mood of the user, user has to classify the songs into multiple emotions and then for playing the songs user has to manually select a particular emotion. According to the emotion, the music will be played from the predefined directories.

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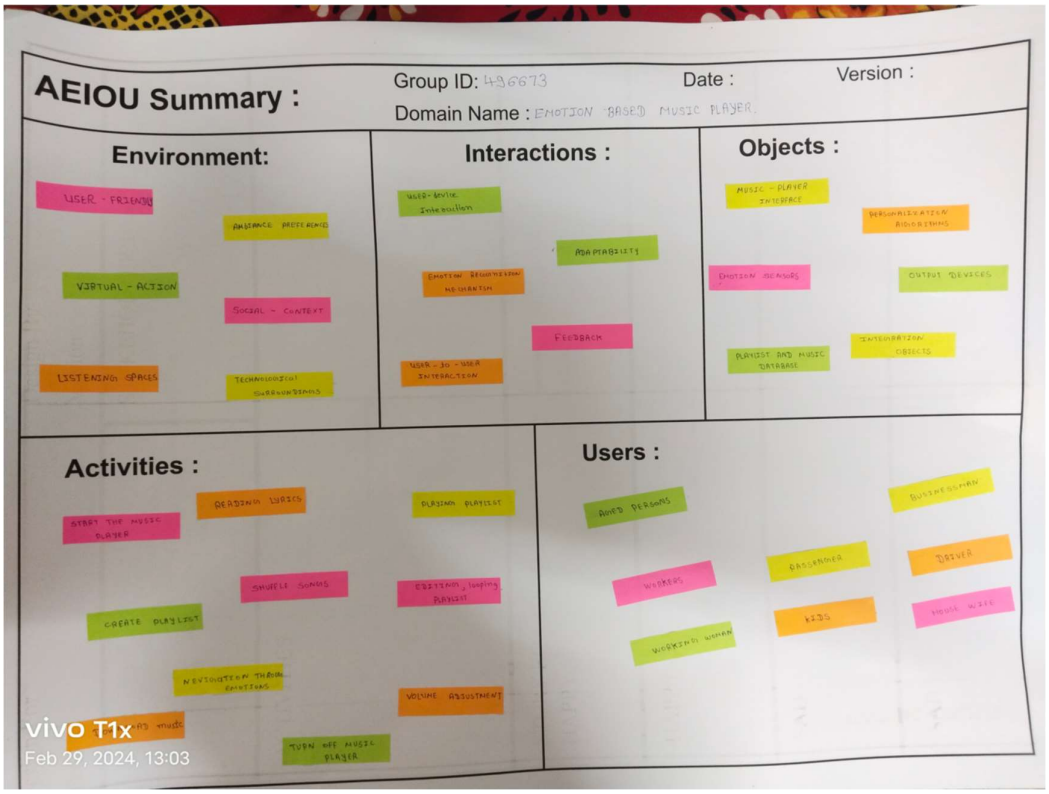
CHAPTER 2

PHASE 1

REVERSE ENGINEERING

2.1 Modification in canvases based on reverse engineering

2.1.1 AEIOU Canvas and description



Summary of AEIOU Canvas and their observations.

ACTIVITIES:- Reading lyrics, Shuffle songs,Create playlist,Start The Music Player, Volume Adjustment

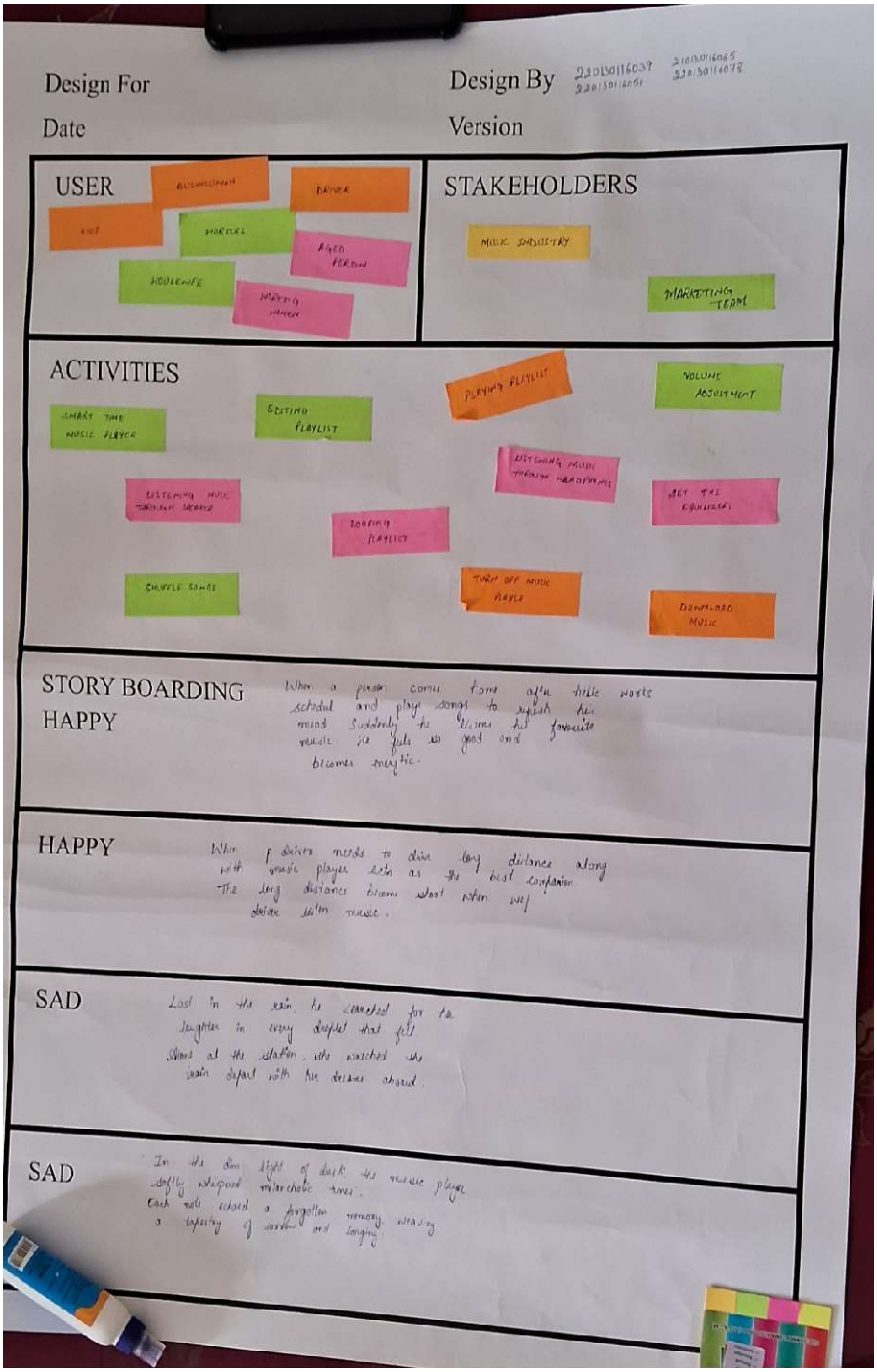
ENVIRONMENT: - User Friendly, Virtual Action, Listening Spaces, Technology Surroundings, Ambiance Preferences

INTERACTIONS: User-device Interaction, Adaptability, Emotion Recongnization Mechnisam, Feedback

OBJECTS: - Music-Player Interface, Personalization Algorithms, Emotion Sensors, Output Devices, Playlist And Music Database

USERS: - Aged Persons, Workers, Kids, Passanger, Businessmen, Driver, Student

2.2.2 Empathy canvas images and description



Summary of Empathy Canvas and their observations.

In empathy canvas, we have covered different areas like who would be the users, who are the stakeholders, what are the activities and what are the story boarding associated with our topic.

USER: - Businessman, driver, Kids, Housewife, Workers, Aged Person, Student

STACKHOLDERS: - Here we have, Music Industry, Marketing team , Researchers, Users, etc.

ACTIVITIES:-

In ACTIVITIES, We Have, Editing playlist , Listening Music Through Speaker, Shuffle Songs, Download Music , Set The Equilizers,etc.

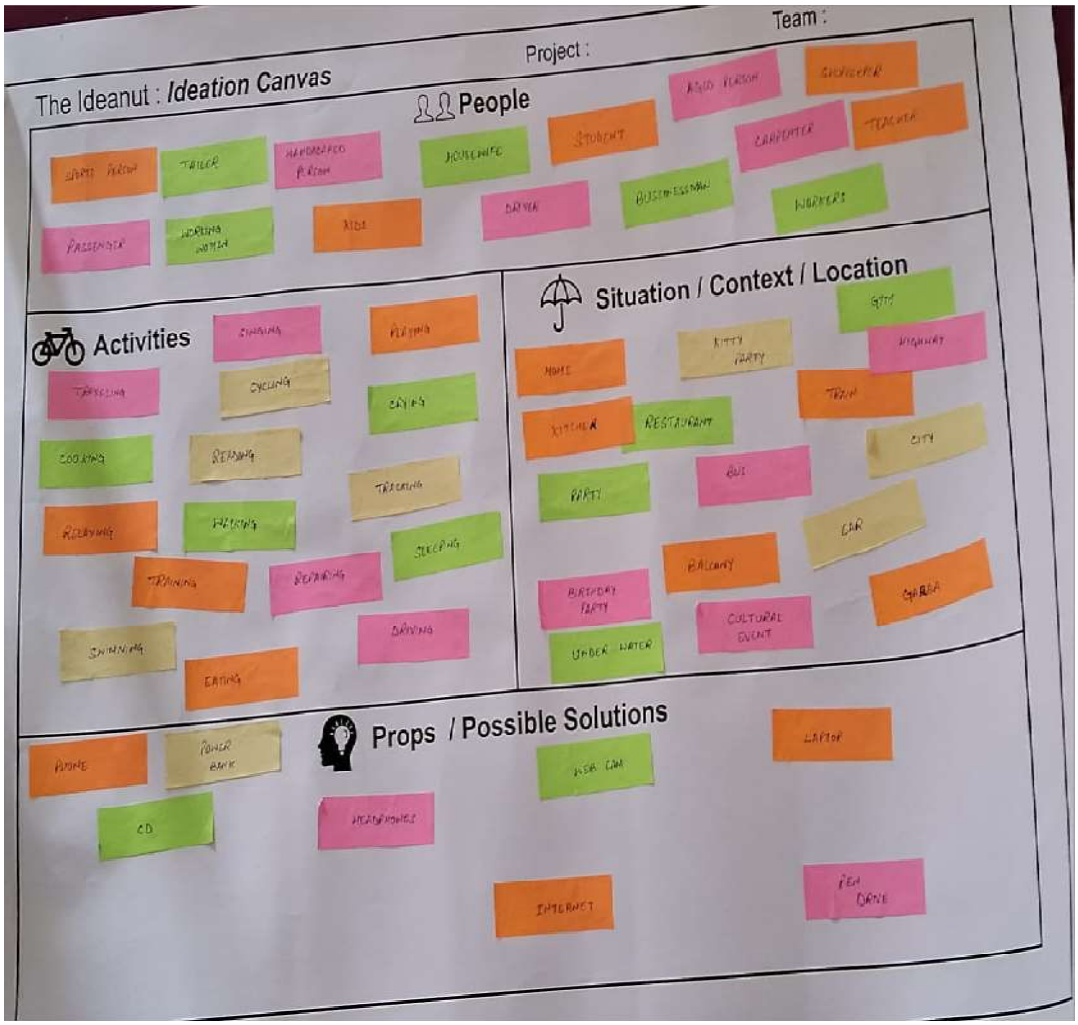


Figure 2.2.3: Ideation Canvas

Summary of Ideation Canvas and their observations.

we started with Ideation Canvas where first we started with people, where we simply thought about the people for whom we want to solve the problem.

For our Product we listed the people like businessman, Working women, Student, kids, Drivers etc. Then list out whatever activity every segment of people do.

Then thought for context/location/situation and finally for possible solutions .We conclude that people's emotions when using an interactive system do not only influence the current interaction process itself (e.g. short-time usage when being bored), but also the likelihood of reusing the system in the future as well as the effectiveness of the application's intention.

So our problem definition which we wanted to develop is Emotion based Music player.

2.2.4 Product development canvas image and description

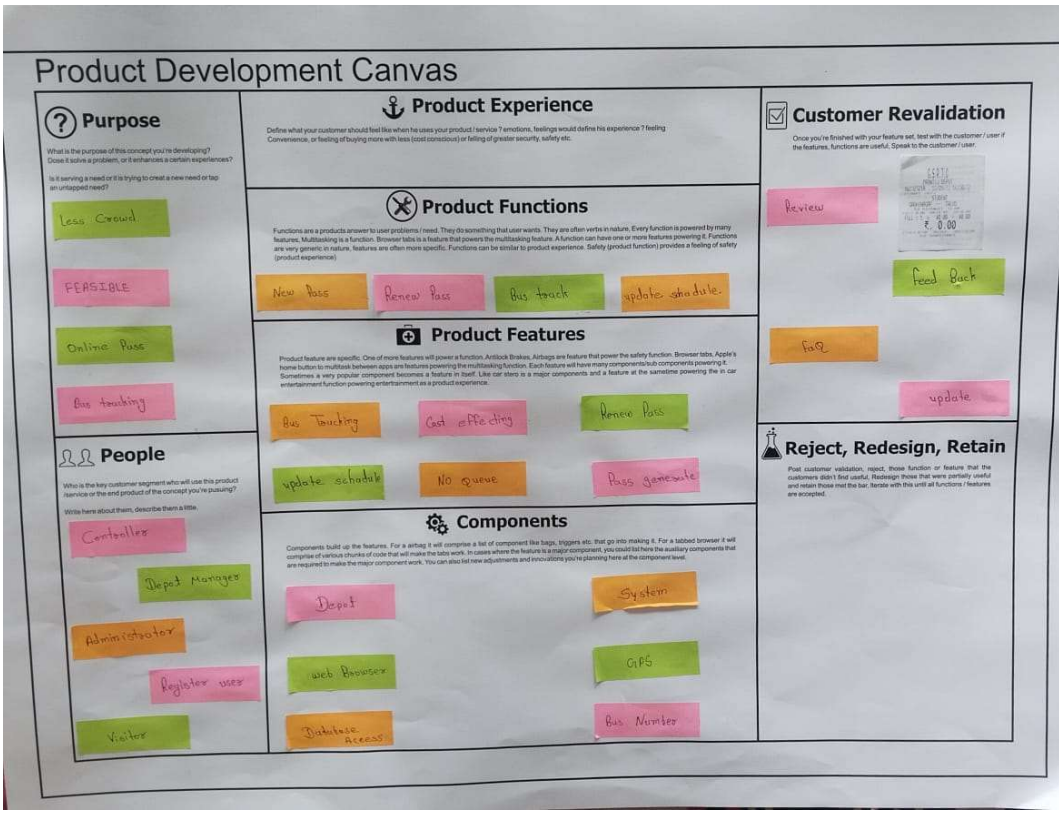
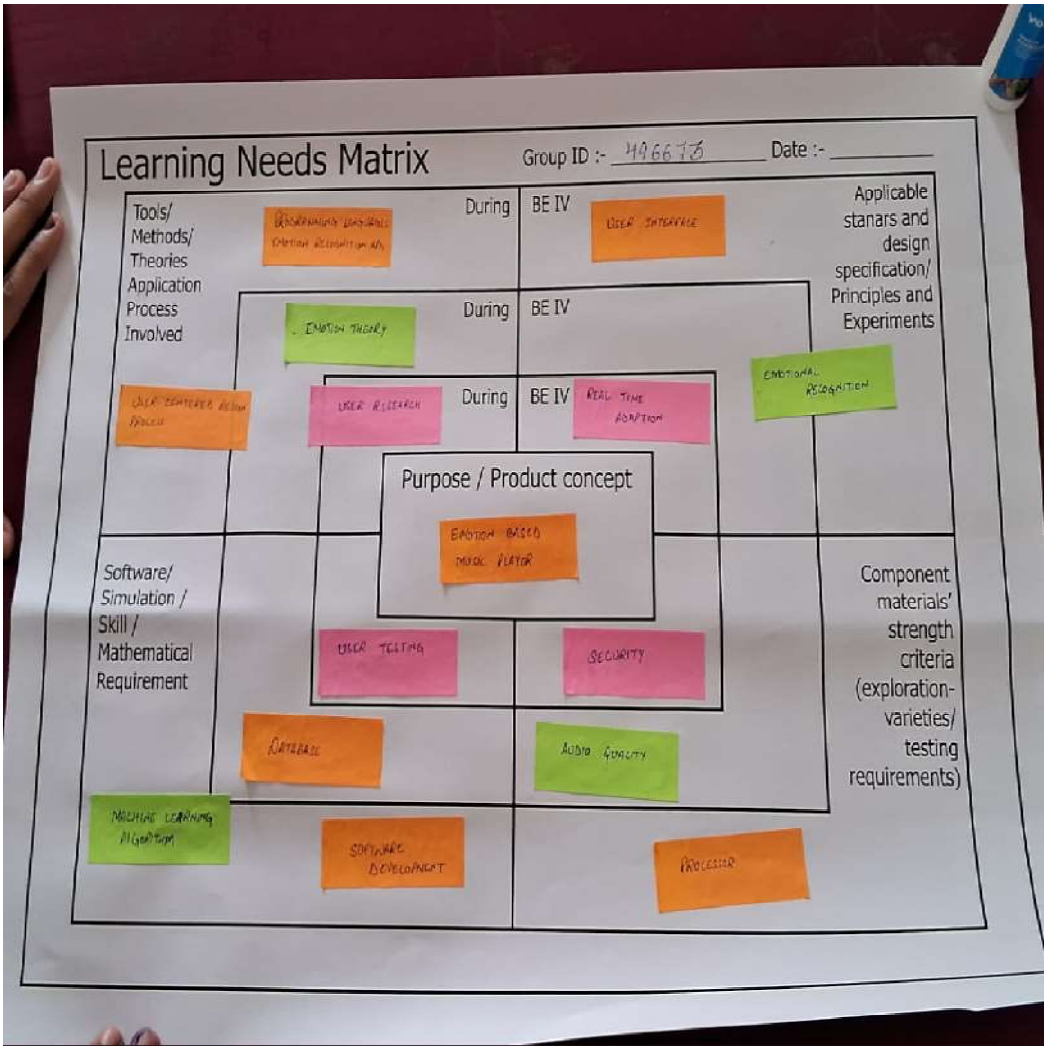


Figure 2.2.4: Product Development Canvas

2.2.5 Learning Needs Metrix



Development Canvas and their observations.

we started with product development Canvas. Where we have mentioned purpose of our product, functions, features, revalidation, product experience.

Purpose of our product was to change people's mood according to their present mood. For example pre-music mood of user is sad now music player identify mood and play a song based on Love songs with a happy ending or Disco/dance so post music mood of user will be hopeful or happy.

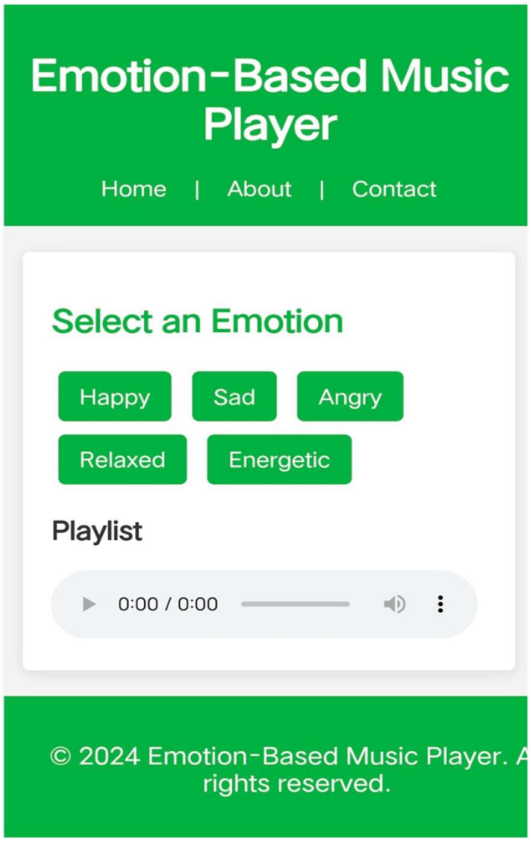
We also visited other teams as a customer for User revalidation.

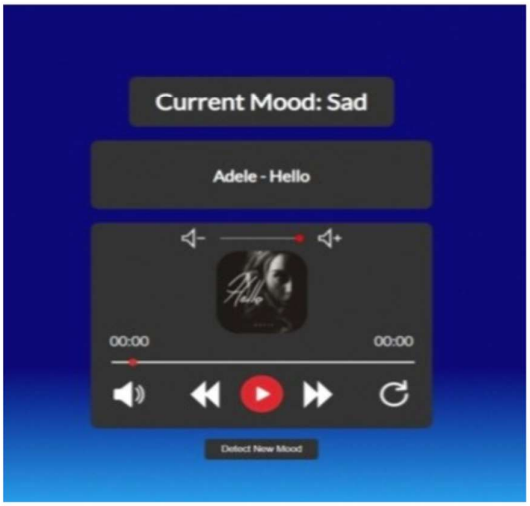
CHAPTER 3

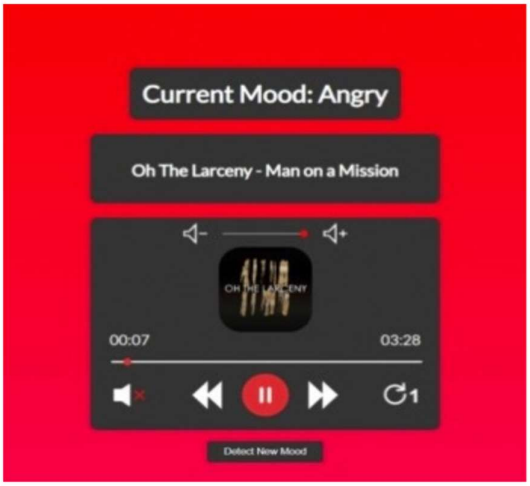
PHASE 2

ROUGH PROTOTYPE MODEL

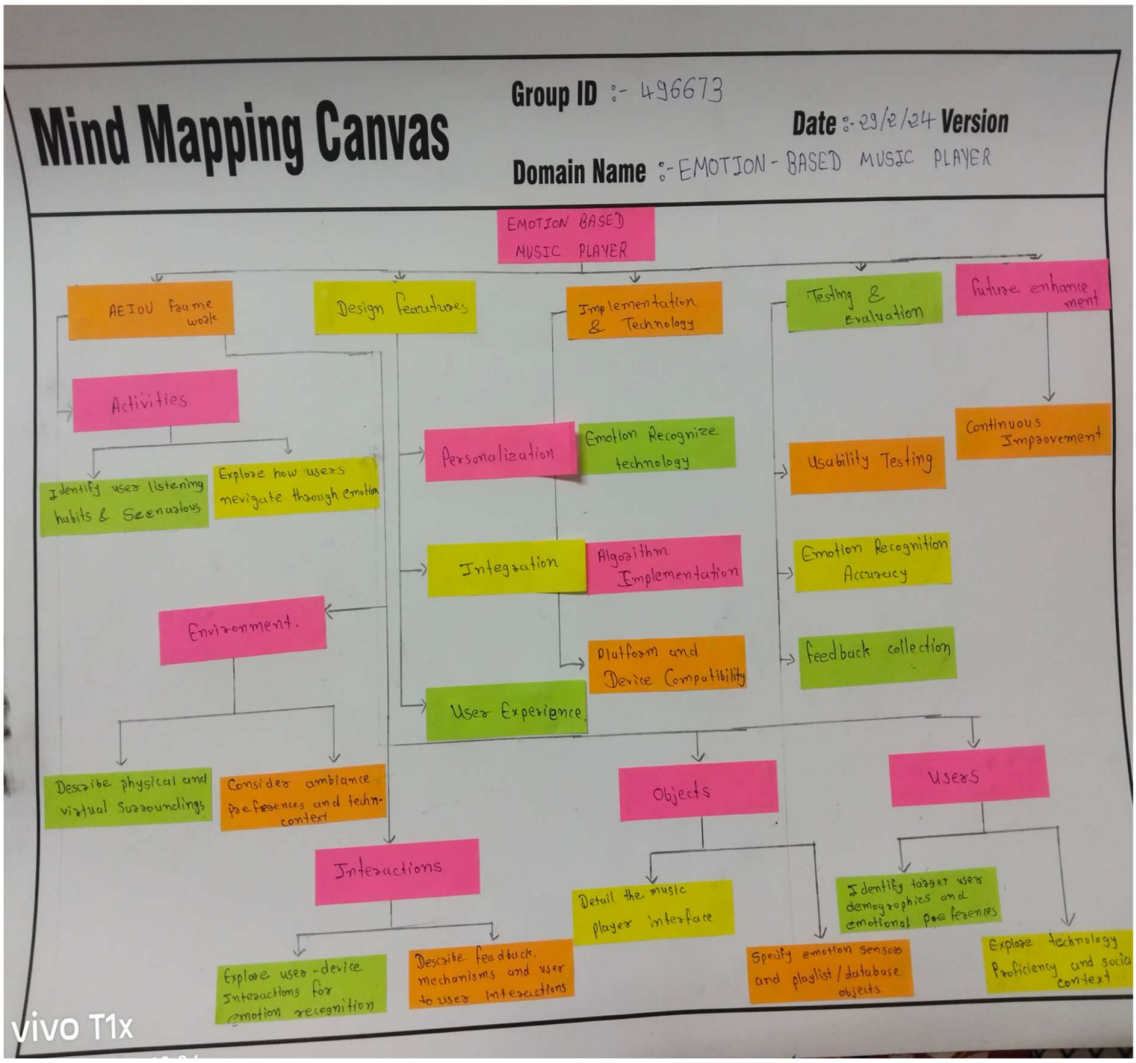
3.1 Prototype Image and Description







4.1 Mind MAP


Mind Mapping Canvas
Group ID :- 496673
Date :- 29/2/24 Version
Domain Name :- EMOTION - BASED MUSIC PLAYER
EMOTION BASED MUSIC PLAYER
AEIOU Frame work
Design features
Implementation & Technology
Testing & Evaluation
Future enhancement
Activities
Identify user listening habits & Seenusions
Explore how users nevigate through emotion
Personalization
Emotion Recognize technology
Integration
Algorithm Implementation
Platform and Device Compatibility
User Experience
Usability Testing
Emotion Recognition Accuracy
feedback collection
Continuous Improvement
Environment.
Describe physical and virtual Surroundings
Consider ambiance Preferences and techn-context
Interactions
Explore user-device Interactions for emotion recognition
Describe feedback mechanisms and user to user interactions
Objects
Detail the music player interface
Specify emotion sensors and playlist / database objects
Users
Identify target users demographics and emotional preferences
Explore technology Proficiency and socio context
vivo T1x

References

[1] https://www.google.com/

[2] https://www.music.com/